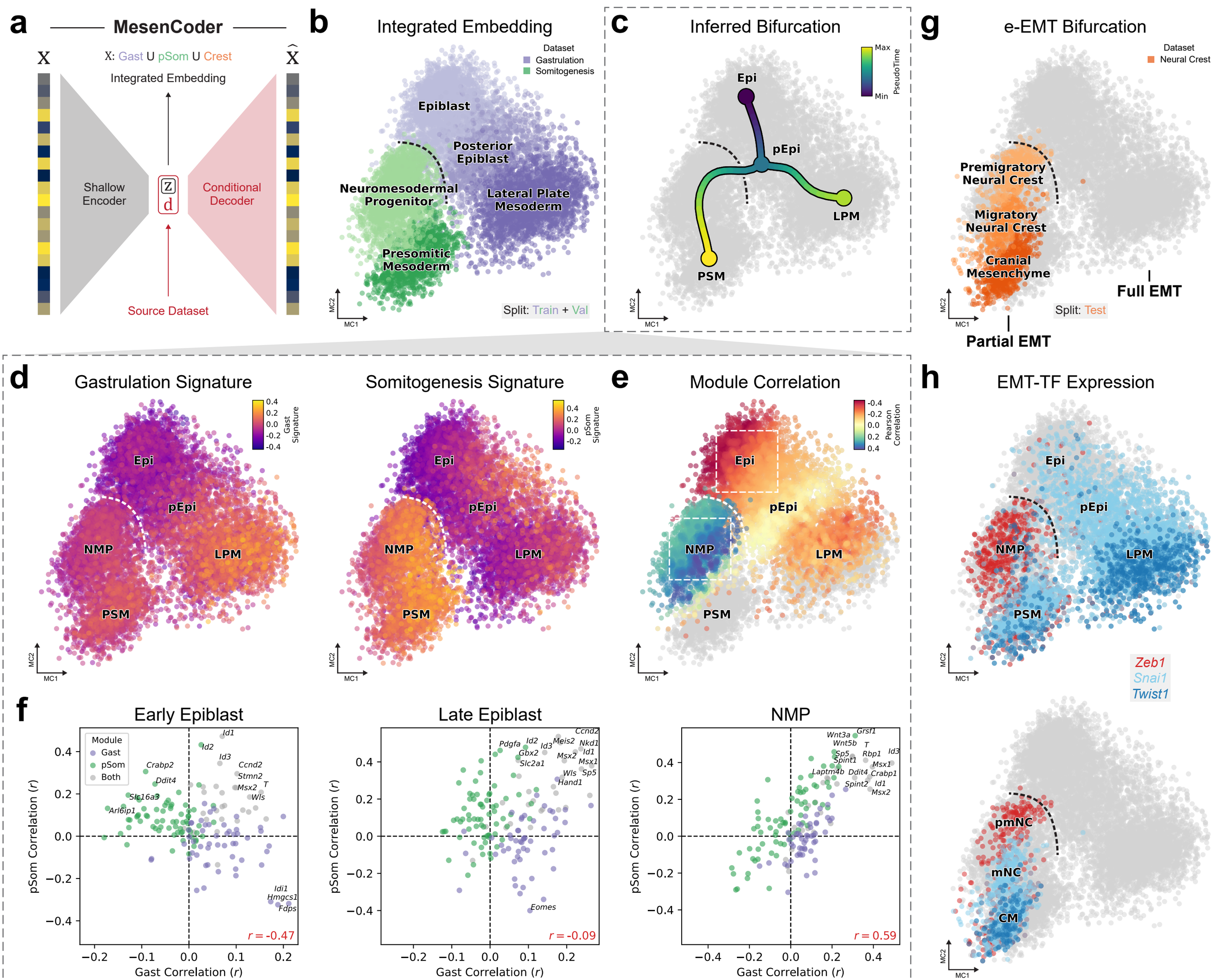

a
MesenCoder
X
X: Gast U pSom U Crest
X̂
Integrated Embedding
Shallow Encoder
z
d
Conditional Decoder
Source Dataset
b
Integrated Embedding
Dataset
Gastrulation
Somitogenesis
Epiblast
Posterior Epiblast
Neuromesodermal Progenitor
Lateral Plate Mesoderm
Presomitic Mesoderm
Split: Train + Val
MC2
MC1
c
Inferred Bifurcation
Max
Min
PseudoTime
Epi
pEpi
LPM
PSM
g
e-EMT Bifurcation
Dataset
Neural Crest
Premigratory Neural Crest
Migratory Neural Crest
Cranial Mesenchyme
Full EMT
Partial EMT
Split: Test
d
Gastrulation Signature
Gast Signature
Epi
pEpi
NMP
LPM
PSM
Somitogenesis Signature
pSom Signature
e
Module Correlation
Pearson Correlation
h
EMT-TF Expression
Zeb1
Snai1
Twist1
f
Early Epiblast
Late Epiblast
NMP
Module
Gast
pSom
Both
pSom Correlation (r)
Gast Correlation (r)
r = -0.47
r = -0.09
r = 0.59
Id1
Id2
Id3
Crabp2
Ddit4
Ccnd2
Stmn2
Msx2
T
Wls
Slc16a3
Arl6ip1
Idi1
Hmgcs1
Fdps
Pdgfa
Gbx2
Slc2a1
Meis2
Nkd1
Msx1
Sp5
Hand1
Eomes
Wnt3a
Grsf1
Wnt5b
Rbp1
Sp5
Spint1
Laptm4b
Ddit4
Crabp1
Spint2
pmNC
mNC
CM